

F5 POC Setup Documentation MultiNIC

[Scope](#)

[Assumption](#)

[Prerequisites](#)

[Deployment Steps](#)

[Phase 1: Prepare the AWS Environment](#)

[Phase 2: Deploy the BIG-IP VE Instance](#)

- [1. Launch Instance](#)
- [2. Attach the Third NIC](#)
- [3. Create an Elastic IP for the BIG-IP Configuration utility](#)
- [4. Add a secondary private IP address for the virtual server](#)
- [5. Create an Elastic IP for the virtual server](#)

[Phase 3: Establish Connectivity Between f5 and the Cloudera application networks](#)

- [1. Create VPC peering between the BIG IP VPC & Application VPC](#)
- [2. Configure Routes in BIG-IP VPC Route Tables \(Toward Cloudera VPC\)](#)
- [3. Configure Routes in Cloudera VPC Route Tables \(Toward BIG-IP VPC\)](#)
- [4. Whitelist F5 VPC in Cloudera security group to allow All traffic](#)
- [5. Test Connectivity](#)

[Phase 4: Internal F5 Configuration](#)

- [1. Set Admin Password](#)
- [2. License and Provision](#)
- [3. Create internal and external VLANs:](#)
- [4. Create internal and external self IPs](#)
- [5. Create Route to access application from external network](#)
- [6. Create a pool and add members to it.](#)
- [7. Steps to Configure F5 Access to different applications](#)
 - [a. Import application server Certificate into F5](#)
 - [b. Create Server SSL Profile](#)
 - [c. Create iRule](#)
 - [d. Cloudera Manager Access – Reverse Proxy / Load Balancer in Front](#)
 - [e. Knox Access – Reverse Proxy / Load Balancer in Front](#)
- [8. Create a virtual server to access different applications](#)
 - [a. Cloudera Manager](#)
 - [b. Hue](#)
 - [c. Knox](#)
 - [d. Nifi](#)
 - [e. ECS Control Plane](#)

[Phase 5: Accessing different applications via F5](#)
[1. Using DNS \(Recommended for Production\)](#)
[2. Using Local Host Mapping \(Testing / POC Only\)](#)
[POC Environment Details](#)

Scope

This document describes how to leverage **F5 BIG-IP** to enable **IPv6 network traffic** to a Cloudera on-premises environment via IPv4 translation. It also outlines the steps required to enable **IPv4 network access** to the same environment. The scope of this document is limited to **Cloudera services that expose REST APIs**.

Assumption

This document assumes that you already have a Cloudera on premises environment deployed in an ipv4 only VPC and you need to enable ipv6 traffic to the existing environment along with ipv4 traffic via F5 BIG IP.

Prerequisites

- Obtain F5 BIG IP VE license
Drop an email to F5 contact person who can provide you with the trial license that has an expiry date of 45 days.
- An IPv6 Enabled VPC

Deployment Steps

This deployment shows three subnets:

- An external, public subnet, where you'll create a virtual server to accept Internet traffic.
- An internal, private subnet, where your application servers live.
- A management subnet, where you can access the BIG-IP Configuration utility; you use the Configuration utility to configure BIG-IP VE.

Traffic flows from clients through BIG-IP VE to application servers.

You create all IP addresses and network interfaces in AWS. Then in BIG-IP VE, you create corresponding objects for the same IP addresses.

Phase 1: Prepare the AWS Environment

- To create a multi-NIC configuration, you must first create an Ipv6 enabled Amazon virtual private cloud (VPC). This is the network environment where your instances will reside.

Note: BIG-IP VPC CIDRs must not overlap with Cloudera VPC CIDRs.

- a. In the AWS Management Console, from the Services menu at the top of the screen, select **VPC**
- b. Click **Create VPC**
- c. Select **VPC only**
- d. Provide **Name tag** e.g f5-bigip
- e. **IPv4 CIDR** e.g 10.0.0.0/16
- f. Under IPv6 CIDR block, Select **Amazon-provided IPv6 CIDR block**

VPC > Your VPCs > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

f5-bigip

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☐ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☒ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (\$) [Info](#)
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode
See which resources in your VPC are unencrypted but allow the creation of unencrypted resources. ☐ Enforce mode
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q f5-bigip	Remove tag

- Before launching your instance, you must establish the networking foundation within your Amazon VPC.
 - Create an SSH Key Pair:** You will need this to access the instance via the command line for initial configuration.
 - Create three subnets in the VPC:** Create subnets with ipv4 and ipv6 cidr in the same availability zone by navigating to **VPC > Subnets > Create subnet**
 - Management Subnet:** Where you can access the BIG-IP Configuration utility(e.g., 10.0.0.0/24, 600:1f16:123:600::/64).
 - External Subnet:** where you'll create a virtual server to accept Internet traffic (e.g., 10.0.1.0/24, 2600:1f16:123:601::/64).
 - Internal Subnet:** where you'll create a virtual server to accept Internet traffic (e.g., 10.0.2.0/24, 2600:1f16:123:602::/64) in the **same Availability Zone** as the first two.

The image below depicts the 1st subnet creation. On the same page you can add 2 more subnets **External Subnet & Internal Subnet**.

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 CIDR block
☒ Manual input ☐ No IPv4 CIDR

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs
< > ^ v

IPv6 CIDR block
☒ Manual input ☐ No IPv6 CIDR

IPv6 VPC CIDR block [Info](#)
- optional
Choose the VPC's IPv6 CIDR block for the subnet. The subnet's IPv6 CIDR must lie within this block.

IPv6 subnet CIDR block
 18.4Q IPs
< > ^ v **Use the arrows to change the CIDR range**

Tags - optional

Key	Value - optional	
Name	Management Subnet	Remove

3. Configure Routing:

- Create an **Internet Gateway** by navigating to **VPC > internet Gateways > Create internet gateway** and attach it to your VPC.

☰ VPC > Internet gateways > Attach to VPC (igw-0362dcf815970eaa)

Attach to VPC (igw-0362dcf815970eaa) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q vpc-01239ef8ed265fe22 X

▶ [AWS Command Line Interface command](#)

[Cancel](#) [Attach internet gateway](#)

- Edit the default route table to add a route for **0.0.0.0/0** and **::/0** targeting the **Internet Gateway**.
- Associate it with the **External & Management Subnet**

☰ VPC > Route tables > rtb-07c473ff552f8a1d6

VPC dashboard <

AWS Global View [L](#)

Filter by VPC: ▼

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables**
- Internet gateways
- Egress-only Internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

▼ **Security**

- Network ACLs
- Security groups

▼ **PrivateLink and Lattice**

rtb-07c473ff552f8a1d6

Details [Info](#)

Route table ID
rtb-07c473ff552f8a1d6

VPC
vpc-02b5e1440196ad656 | f5-bigip

Main
Yes

Owner ID
037356316600

Routes Subnet associations Edge associations Route propagation Tags

Routes (4)

Q Filter routes

Destination	Target	Status
::/0	igw-06c02c9e54511bba4	✓ Active
2600:1f16:123:600::/56	local	✓ Active
0.0.0.0/0	igw-06c02c9e54511bba4	✓ Active
10.0.0.0/16	local	✓ Active

rtb-07c473ff552f8a1d6

Details Info

Route table ID

rtb-07c473ff552f8a1d6

Main

Yes

Explicit subnet associations

2 subnets

Edge associations

-

VPC

vpc-02b5e1440196ad656 | f5-bigip

Owner ID

037356316600

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ipv6-external	subnet-0eeae11712933c4ed	10.0.1.0/24	2600:1f16:123:601::/64
ipv6-management	subnet-0d1d968c57f841b1c	10.0.0.0/24	2600:1f16:123:600::/64

4. **Set Up Security Groups:** Create three groups associated with the vpc to control traffic by navigating to VPC > Security Groups > Create security group:

- **Management:** Allows SSH (22) and HTTPS (443) from your specific IP.
- **VirtualServer:** Allows HTTP/HTTPS (80/443) for clients.
- **InternalTraffic:** Allows TCP ports like 4353, 6699 and UDP port 1026 for BIG-IP config sync and failover.

VPC > Security Groups

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Security Groups (3) Info

Find security groups by attribute or tag

vpc-015a8e45111312518 X Clear filters

Name	Security group ID	Security group name
-	sg-0200293ef5d52d337	Management
-	sg-08e82a062c2e328ba	VirtualServer
-	sg-0f21caa5fe70a1515	default

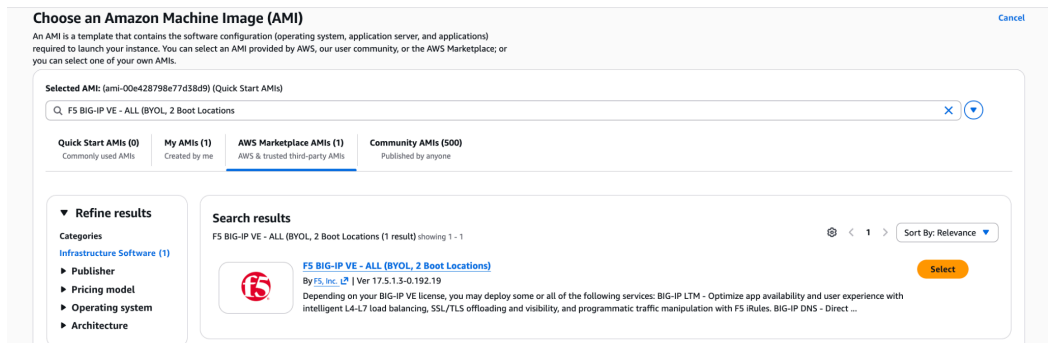
Phase 2: Deploy the BIG-IP VE Instance

With the network ready, you can now launch the F5 software from the AWS Marketplace.

1. Launch Instance

Use “F5 BIG-IP VE - ALL (BYOL, 2 Boot Locations)” AMI.

- In the AWS Management Console, from the Services menu at the top of the screen, select **EC2**
- Click **Launch instance**
- Click **Browse more AMIs**
- Search for “**F5 BIG-IP VE - ALL (BYOL, 2 Boot Locations)**” under the **AWS Marketplace AMIs**
- Select** and Click **Subscribe Now**



- Provide **Name**, Select **key pair**.
- Edit **Network settings**
 - Under **Network**, Select the f5 enabled VPC
 - Under **Subnet**, Select **Management Subnet**.
 - For security groups, choose **Select existing security group** and select **Management**, **VirtualServer**, **InternalTraffic** security groups from the dropdown.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

f5poc-akt ▼

[Create new key pair](#)

▼ Network settings [Info](#)

VPC - *required* [Info](#)

vpc-02b5e1440196ad656 (f5-bigip) ▼

10.0.0.0/16 2600:1f16:123:600::/56

Subnet [Info](#)

subnet-0d1d968c57f841b1c

ipv6-management

VPC: vpc-02b5e1440196ad656 Owner: 037356316600 Availability Zone: us-east-2a (use2-az1) ▼

Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.0.0/24

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable ▼

Auto-assign IPv6 IP [Info](#)

Disable ▼

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group
 ☒ Select existing security group

Common security groups [Info](#)

Select security groups ▼

ipv6new-management sg-0ee89df3d0daad58e ✕

VPC: vpc-02b5e1440196ad656

ipv6new-internaltraffic sg-0c8673488253d8514 ✕

VPC: vpc-02b5e1440196ad656

ipv6new-virtualserver sg-0e5b98e8e435bd080 ✕

VPC: vpc-02b5e1440196ad656

[Compare security group rules](#)

- h. Click on **Advanced Network Configuration**
 - i. Add primary ip e.g **10.0.0.200** and for IPv6 IPs, select **manually assign** and provide value e.g **2600:1f16:123:600::200**
 - ii. Click on **Add Network Interface**

▼ Advanced network configuration

Network interface 1

Device index | [Info](#)
0

Subnet | [Info](#)
subnet-Od1d968c57f841b1c
IP addresses available: 250

Primary IP | [Info](#)
10.0.0.200

Secondary IP | [Info](#)
Select

IPv6 IPs | [Info](#)
Manually assign
2600:1f16:123:600::200 ✕

IPv4 Prefixes | [Info](#)
Select

IPv6 Prefixes | [Info](#)
Select

Assign Primary IPv6 IP | [Info](#)
Yes

Delete on termination | [Info](#)
Select

Interface type | [Info](#)
Select

Network card index | [Info](#)
Select
The selected instance type does not support multiple network cards.

ENA Express | [Info](#)
Select
The selected instance type does not support ENA Express.

ENA Express UDP | [Info](#)
Select
The selected instance type does not support ENA Express.

ENA queues | [Info](#)
The selected instance type does not support ENA queues.

Idle connection tracking timeout | [Info](#)
☐ Enable

[Add network interface](#)

- i. Similarly for 2nd interface change Subnet to **External Subnet**, add primary ip e.g **10.0.1.200** and for IPv6 IPs, select **manually assign** and provide value e.g **2600:1f16:123:601::200**

Network interface 2 Remove

Device index [Info](#)
1

Subnet [Info](#)
subnet-0d1d968c57f841b1c
IP addresses available: 250

Primary IP [Info](#)
10.0.1.200

Network interface [Info](#)
New interface

Security groups [Info](#)
Select security groups
[Show all selected \(3\)](#)

Secondary IP [Info](#)
Select

Description [Info](#)

Auto-assign public IP [Info](#)
Select

IPv6 IPs [Info](#)
Manually assign
2600:1f16:123:601::20

IPv4 Prefixes [Info](#)
Select

IPv6 Prefixes [Info](#)
Select

Assign Primary IPv6 IP [Info](#)
Select

Delete on termination [Info](#)
Select

Interface type [Info](#)
Select

Network card index [Info](#)
Select
The selected instance type does not support multiple network cards.

ENA Express [Info](#)
Select
The selected instance type does not support ENA Express.

ENA Express UDP [Info](#)
Select
The selected instance type does not support ENA Express.

ENA queues [Info](#)
The selected instance type does not support ENA queues.

Idle connection tracking timeout [Info](#)
☐ Enable

- j. Review the details and Click **Launch Instance**.

2. Attach the Third NIC

When you launched the BIG-IP VE instance in the previous step, you associated two network interfaces with it (one for management and one for external). To connect BIG-IP VE with your backend application servers, create an internal network interface, and attach it to your BIG-IP VE instance.

- In the AWS Management Console, from the Services menu at the top of the screen, select **EC2**.
- In the Navigation pane, under NETWORK & SECURITY, select **Network Interfaces**.
- Enter names for your existing network interfaces. In the list, there should be two interfaces for BIG-IP VE. Use the values in the **IPv6 addresses(Sort with)**

column to determine which NIC is for which subnet.

	Name		Network interface ID	Subnet ID
☒	management		eni-0eaf3e60ec126ec48	subnet-0a04f9dc10b7875d9
☐	external		eni-0c5d5c9a7549df0ca	subnet-061af078161896e52

d. Click **Create Network Interface** and populate the appropriate field

Field	Value
Description	Internal
Subnet	10.0.2.0./24(internal)
Private IPv4	10.0.2.200
Security groups	InternalTraffic

EC2 > Network Interfaces > eni-0a6e41f56f3d5cab9 > Manage IP addresses

Manage IP addresses

Assign or unassign IPv4 and IPv6 addresses to or from a network interface.

IP addresses

To assign additional public IPv4 addresses to this network interface, you must [allocate](#) Elastic IP addresses and associate them with this network interfaces.

▼ eth2: eni-0a6e41f56f3d5cab9 - Internal - 10.0.2.0/24

IPv4 addresses

Private IP address: 10.0.2.200 Public IP address:

IPv6 addresses

No IP addresses present

Assign primary IPv6 IP

☐ Enable

Makes the first IPv6 address that is assigned to the network interface the primary IPv6. If this is a primary network interface, once a primary IPv6 address is assigned, it cannot be unassigned.

A minimum of 1 IPv6 IPs must be assigned to the network interface.

Auto-assign public IP

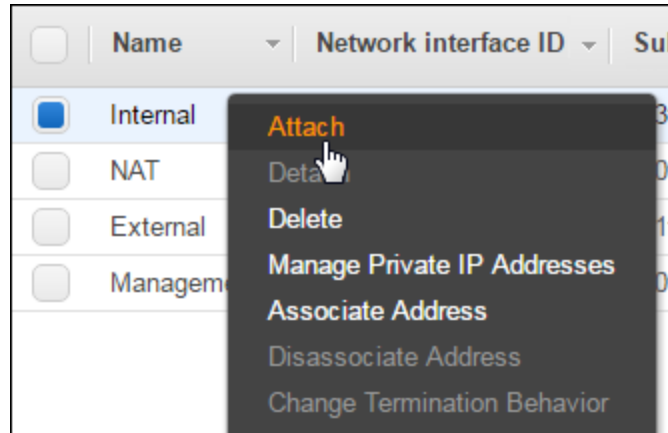
☐ Allow

Allows you to reassign a private IPv4 address that is assigned to another network interface or instance to this network interface.

☐ Allow

e. Attach the newly created interface to the BIG IP instance.

- Update the name in the list to **Internal**.
- Select the new **Internal** network interface, Click Actions and select Attach.



- f. Select the f5 VPC, big ip instance and click **Attach**

Attach network interface

Network interface
eni-0a6e41f56f3d5cab9 (Internal)

VPC

vpc-01239ef8ed265fe22

Instance

i-068ec1bcc7cc1403d

▼ **Advanced ENA Configuration - optional**

ENA Configuration

The selected instance type (m5.xlarge) does not support the following feature(s):

- ENA Express and ENA Express UDP
- ENA queues

ENA Express
Enable ENA Express to increase the maximum bandwidth used per stream and minimize tail latency of network traffic between EC2 instances.
☐ Enable

ENA Express UDP
Use ENA Express for UDP traffic.
☐ Enable

ENA queues
8 (default)
Min: 1, Max: 2. The value must be a power of 2. Available ENA queues across all ENIs: -16.

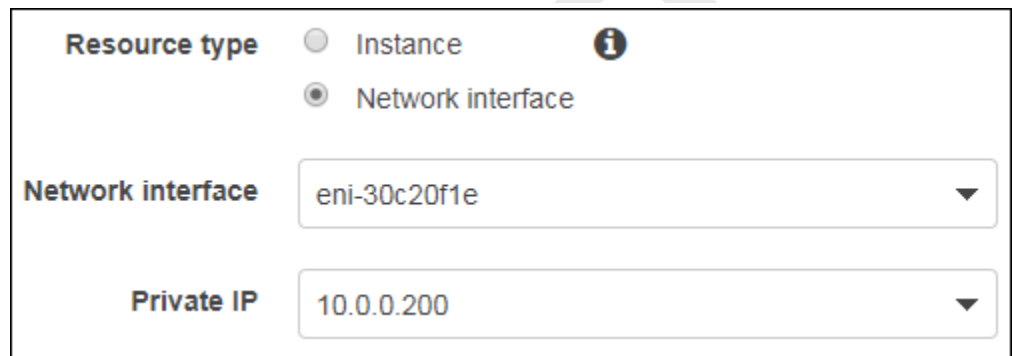
Cancel Attach

- g. **Reboot the BIG-IP VE** immediately after attaching the third NIC so the system can register the new interface.

3. Create an Elastic IP for the BIG-IP Configuration utility

In order to access the BIG-IP Configuration utility from the Internet, the BIG-IP VE instance must have an Elastic IP (EIP) address associated with it.

- In the AWS Management Console, from the Services menu at the top of the screen, select EC2.
- In the Navigation pane, under NETWORK & SECURITY, select Elastic IPs.
- Click **Allocate new address**.
- Click **Allocate**.
- Select the newly created EIP, click **Actions** and select **Associate address** from the popup menu screen
- For Resource type, select **Network interface**, for Network interface, select the **management** NIC, and for Private IP select the management subnet's private IP address, **10.0.0.200**.



- Click **Associate**.

Later, you will use this EIP to access the BIG-IP Configuration utility.

4. Add a secondary private IP address for the virtual server

Before you can create a virtual server, you must assign a secondary IP address to the external network interface. You will associate this secondary IP address with the BIG-IP VE virtual server address.

- In the AWS Management Console, from the Services menu at the top of the screen, select **EC2**.
- In the Navigation pane, under NETWORK & SECURITY, select Network Interfaces.
- Identify the external network interface (the NIC that uses the **2600:1f16:123:601::200** ipv6 ip).
- Select the **external** network interface, click **Actions** and click **Manage IP Addresses**.
- Below the existing address, select **Assign new IP address**
- Type ipv4 address e.g 10.0.1.202 and ipv6 address e.g 2600:1f16:123:601::202
- Click **Save**.

Manage IP addresses Info
Assign or unassign IPv4 and IPv6 addresses to or from a network interface.

IP addresses

To assign additional public IPv4 addresses to this network interface, you must [allocate](#) Elastic IP addresses and associate them with this network interfaces.

▼ eth1: eni-0c9839e72f4cb8487 - 10.0.1.0/24

IPv4 addresses

Private IP address	Public IP address	
10.0.1.166		<button>Unassign</button>
10.0.1.202		<button>Unassign</button>

Assign new IP address

IPv6 addresses

IPv6 address	
2600:1f16:123:601::200	<button>Unassign</button>
2600:1f16:123:601::202	<button>Unassign</button>

Assign new IP address

Assign primary IPv6 IP Info
Makes the first IPv6 address that is assigned to the network interface the primary IPv6. If this is a primary network interface, once a primary IPv6 address is assigned, it cannot be unassigned.

☐ Enable

Auto-assign public IP Info

☒

Allow secondary private IPv4 addresses to be reassigned
Allows you to reassign a private IPv4 address that is assigned to another network interface or instance to this network interface.

☐ Allow

Cancel Save

5. Create an Elastic IP for the virtual server

(Follow this step specifically for ipv4 to ipv4 routing)

Before you begin, note the secondary private IP address assigned to the **external** network interface. If you followed the steps in the previous task, the secondary private IPv4 address is **10.0.1.202**.

Now make the virtual server IP address (the secondary private IP address) accessible to the Internet by associating an Elastic IP (EIP) address with it.

- In the AWS Management Console, from the Services menu at the top of the screen, select **EC2**.
- In the Navigation pane, under NETWORK & SECURITY, select **Elastic IPs**.
- Click **Allocate new address**.
- Click **Allocate**, and then click **Close**.
- From the list of EIP addresses, Select the newly created address, click **Actions** and select **Associate address** from the popup menu.
- Select the **external** network interface and the secondary IP address (10.0.1.202).
- Click **Associate**.

Phase 3: Establish Connectivity Between f5 and the Cloudera Environment networks

The F5 machine should be able to communicate with the Cloudera on premises environment. If the Cloudera environment is already deployed in an ipv4 network, F5 BIG IP VE can be deployed in an IPv6 enabled VPC and connectivity can be established via VPC peering.

1. Create VPC peering between the BIG IP VPC & Application VPC

- Go to **VPC** → **Peering Connections**
- Click **Create peering connection**
- Configure:
 - **Name tag:** `bigip-to-cloudera-peering`
 - **VPC (Requester):** BIG-IP VPC (e.g f5-bigip)
 - **Account:** Same account or Another account
 - **Region:** Same region
 - **VPC (Acceptor):** Cloudera/Application VPC

VPC > Peering connections > Create peering connection

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

Select a local VPC to peer with

VPC ID (Requester)

VPC CIDRs for vpc-02b5e1440196ad656 (f5-bigip)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-
2600:1f16:123:600::/56	Associated	-

Select another VPC to peer with

Account

☒ My account
☐ Another account

Region

☒ This Region (us-east-2)
☐ Another Region

VPC ID (Acceptor)

VPC CIDRs for vpc-0173aec9a046304f7 (cdpar-725-pvc-base)

CIDR	Status	Status reason
10.10.0.0/16	Associated	-

- Click **Create peering connection**

e. Accept Peering

- Go to **VPC → Peering Connections**
- Select the same pending peering connection
- Click **Actions → Accept request**
- Verify status is **Active**

Peering connections (1/2) [Info](#)

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC
bigip-to-cloudera-peering	pcx-057bb13bc77f14a5f	Active	vpc-02b5e1440196ad656 / f5-bigip

2. Configure Routes in BIG-IP VPC Route Tables (Toward Cloudera VPC)

Traffic from BIG-IP to Cloudera must be routed via the peering connection.

- Create a new route table, goto **VPC > Route Tables > Create route table**
- Provide **Name** and your f5 **VPC**
- Click **Create route table**

[VPC](#) > [Route tables](#) > [Create route table](#)

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

VPC
The VPC to use for this route table.

- Click **Edit routes → Add route**
- Add destination as **Cloudera VPC IPv4 CIDR** and target as **peering connection**.

rtb-05ff885f9c4963732 / f5vpc-to-cdppvpc

Details [Info](#)

Route table ID

 rtb-05ff885f9c4963732

VPC

[vpc-02b5e1440196ad656](#) | [f5-bigip](#)

Main

 No

Owner ID

 037356316600

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

 *Filter routes*

Destination	Target	Status
2600:1f16:123:600::/56	local	Active
10.0.0.0/16	local	Active
10.10.0.0/16	pcx-057bb13bc77f14a5f	Active

- f. Click **Subnet associations** and Associate **Internal** Subnet.

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (1)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ipv6-internal	subnet-044897bf161545a8e	10.0.2.0/24	2600:1f16:123:602::/64

3. Configure Routes in Cloudera VPC Route Tables (Toward BIG-IP VPC)

Return traffic from Cloudera to BIG-IP must also traverse the peering connection.

- a. Go to **VPC → Route Tables**

- b. Select route tables associated with Cloudera private subnets/Application subnets
- c. Click **Edit routes → Add route**
- d. Add destination as **F5 VPC IPv4 CIDR** and **IPv6 CIDR** and target as **peering connection**.

rtb-0cf52ef931634b849 / cdpar-725-pvc-base-private-01

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (4)

Q

Filter routes

Destination	Target	Status
2600:1f16:123:600::/56	pcx-057bb13bc77f14a5f	Active
10.10.0.0/16	local	Active
10.0.0.0/16	pcx-057bb13bc77f14a5f	Active

4. Whitelist F5 VPC in Cloudera security group to allow All traffic

- a. Whitelist IPv4 CIDR (e.g 10.0.0.0/16)
- b. Whitelist IPv6 CIDR (e.g 2600/1f18.307d.8200::/56)

EC2 > Security Groups > sg-0a680d72a82c584d4 - pocf5-pvc-base-intra > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	
sg-0cd4e0077b3a7c1a5	All traffic	All	All	Custom	Self-reference ingress rule	Delete
sg-0475813e9e9f4cae	All traffic	All	All	Custom	sg-0a680d72a82c584d4	Delete
sg-0174749076845633a	All traffic	All	All	Custom	10.0.0.0/16	Delete
					2600:1f18:307d:8200::/56	Delete

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

5. Test Connectivity

- a. SSH to the f5 server using management interface ip and keypair
- b. Enter **bash** command to switch bash shell
- c. Try to curl Cloudera Application from the F5 machine

Shell

```
curl -vk https://<App serverip>:7183/cm
```

You should get a response as below which confirms connectivity.

```
P[admin@i-0ed1e144a2fc859e8:Active:Standalone] ~ # curl -v https://10.10.1.159:7183/cmfs -k
* Trying 10.10.1.159...
* Connected to 10.10.1.159 (10.10.1.159) port 7183 (#0)
* ALPN, offering h2
* ALPN, offering http/1.1
* Cipher selection: ALL:!EXPORT:!EXPORT40:!EXPORT56:!aNULL:!LOW:!RC4:@STRENGTH
* successfully set certificate verify locations:
*   CAfile: /etc/pki/tls/certs/ca-bundle.crt
*   CApath: none
* TLSv1.2 (OUT) TLS header: Certificate Status (22):
```

Phase 4: Internal F5 Configuration

Once the AWS side is complete, configure BIG-IP VE so that traffic passes through it to your application servers.

1. Set Admin Password

Connect via SSH using elastic ip and keypair and use the below command to set a secure password.

Shell

```
modify auth password admin
```

2. License and Provision

Log into the web utility (<https://<EIP>>) and activate your license.

- a. Goto **License**
- b. On the General Properties page, click **Activate**.
- c. In the **Base Registration key** field, enter the case-sensitive registration key received from F5.
- d. Click **Next > Accept agreement**
It'll take some time and the license will be activated.
- e. Under "Resource Provisioning," click **Next > Next > Next**
- f. In the **Advanced Network Configuration** area, click **Finished**.

3. Create internal and external VLANs:

In BIG-IP VE, you must create an **external** and **internal** VLAN that corresponds to the AWS VPC subnets.

- a. On the Main tab, click Network -> VLANs.
- b. Click Create and populate the appropriate fields for the **external** VLAN.

Field	Value
Name	external
Interface	1.1
Tagging	Untagged

- c. Click **Add** and click **Repeat**.

General Properties

Name	external
Description	
Tag	

Resources

Interfaces	Interface: ▼
	Tagging: Untagged ▼
	Add
	1.1 (untagged)

Edit Delete

Configuration: Basic ▼

Source Check	☐
MTU	1500

sFlow

Polling Interval	Default ▼
Sampling Rate	Default ▼

Cancel **Repeat** Finished

- d. Now populate the appropriate fields for the **internal** VLAN.

Field	Value
Name	internal
Interface	1.2
Tagging	Untagged

- e. Click **Finished**.

The screen refreshes, and the two new VLANs are in the list.

Network » VLANs : VLAN List



VLAN List

VLAN Groups

☒	▲ Name	↕ Application	↕ Tag	Untagged Interfaces
☐	external		4094	1.1
☐	internal		4093	1.2

4. Create internal and external self IPs

Before starting these steps, in AWS, note the primary private IPv4 addresses for the external network interface (device index 1) and the internal network interface (device index 2).

Then in BIG-IP VE, create an external and internal self IP address, based on these private IP addresses.

- a. In the BIG-IP VE Configuration utility, on the Main tab, click **Network -> Self IPs**.
- b. Click **Create** and populate the appropriate fields for the external self IP address.

Field	Value
Name	ExternalSelfIP
IP Address	10.0.1.200
Netmask	255.255.255.0
VLAN/Tunnel	external
Port Lockdown	Allow All

Network » Self IPs » ExternalSelfIP



Properties

Configuration

Name	ExternalSelfIP
Partition / Path	Common
IP Address	10.0.1.200
Netmask	255.255.255.0
VLAN / Tunnel	external
Port Lockdown	Allow All
Traffic Group	☐ Inherit traffic group from current partition / path traffic-group-local-only (non-floating)
Service Policy	None

- c. Click **Repeat** and populate the appropriate fields for the internal self IP address.

Field	Value
Name	InternalSelfIP
IP Address	10.0.2.200
Netmask	255.255.255.0
VLAN/Tunnel	internal
Port Lockdown	Allow All

Network » Self IPs » InternalSelfIP

⚙️ Properties

Configuration

Name	InternalSelfIP
Partition / Path	Common
IP Address	10.0.2.200
Netmask	255.255.255.0
VLAN / Tunnel	internal ▾
Port Lockdown	Allow All ▾
Traffic Group	☐ Inherit traffic group from current partition / path traffic-group-local-only (non-floating) ▾
Service Policy	None ▾

d. Click Finished.

The screen refreshes, and the two new self IP addresses are in the list.

☒	Name	Application	IP Address	Netmask	VLAN / Tunnel
☐	ExternalSelfIP		10.0.1.200	255.255.255.0	external
☐	InternalSelfIP		10.0.2.200	255.255.255.0	internal

5. Create Route to access application from external network

- On the Main tab, click **Network -> Routes**.
- Click **Add** and populate the appropriate fields for the route

Field	Value
Name	cdp_pool_route
Destination	10.10.1.0 (network add of cloudera application)

Netmask	255.255.255.0
Gateway Address	10.0.2.1 (gateway ip of internal N/w Interface)

Network » Routes » **cm_pool_route**

⚙️ Properties

Properties

Name	cm_pool_route	
Partition / Path	Common	
Description		
Destination	10.10.1.0	
Netmask	255.255.255.0	
Resource	Use Gateway... ▾	
Gateway Address	IP Address ▾	10.0.2.1
MTU		

Update Delete

6. Create a pool and add members to it.

Traffic goes through BIG-IP VE to a pool. Your application servers should be members of this pool.

- Open a web browser and go to the BIG-IP Configuration utility, using the public IP address on the management network, for example: <https://<external-ip-address>>.
- On the Main tab, click **Local Traffic -> Pools**.
- Click Create.
- In the Name field, type pool name e.g **cm_pool**, **nifi_pool**, **knox_pool**, **ecs_control plane** etc. Names must begin with a letter, be fewer than 63 characters, and can contain only letters, numbers, and the underscore (_) character.
- For Health Monitors, move **https** from the Available to the Active list.

- f. In the New Members section, in the Address field, type the IP address of the application server(CM, Knox, NiFi etc.) .
- g. In the Service Port field, type a service port, for example 7183 (Cloudera Manager), 8443 (Knox) etc.
- h. Click **Add**.

Local Traffic » Pools : Pool List » New Pool...

Configuration: Basic ▾

Name: cm_pool

Description:

Health Monitors

Active		Available
/Common		http2
https	<<	http2_head_f5
	>>	http_head_f5
		https_443
		https_head_f5

Resources

Load Balancing Method: Round Robin ▾

Priority Group Activation: Disabled ▾

New Members

☒ New Node ☐ New FQDN Node ☐ Node List

Node Name: (Optional)

Address: 10.10.1.159

Service Port: 7183 Select... ▾

Add

Node Name	Address/FQDN	Service Port	Auto Populate	Priority
No members to display.				

Edit Delete

Cancel **Repeat** Finished

- i. Click **Repeat** for adding more pool members(CM, Knox, Nifi, Hue etc.) as needed and click **Finished** once done.

- j. For Accessing ECS_Control Plane and data services (Data Engineering, Data Warehouse, Cloudera AI) directly, you can create ECS master nodes as a pool member.

Local Traffic » Pools : Pool List » **ecs_pool**

⚙️ Properties **Members** Statistics ↗

Load Balancing

Load Balancing Method: Round Robin ▾

Priority Group Activation: Disabled ▾

Update

Current Members

☒	▼ Status	Member	▲ Address	Service Port	FQDN	Ephem
☐	●	10.10.1.126:443	10.10.1.126	443		No

Enable Disable Force Offline Remove

The list now contains the pool members created.

Main Help About

Local Traffic » **Pools : Pool List**

⚙️ Pool List Statistics ↗

* Search

☒	▼ Status	▲ Name
☐	●	cm_pool
☐	●	dataviz_pool
☐	●	ecs_pool
☐	●	hue_pool
☐	●	knox_pool
☐	●	nifi_pool

Delete...

Statistics
iApps
DNS
Local Traffic
Network Map
Virtual Servers
Policies
Profiles
Ciphers
iRules
Pools
Nodes

7. Steps to Configure F5 Access to different applications

Some of the Cloudera applications require some preconfigurations before creating the virtual server. Below are the two validated examples for **Knox and NiFi**, More services might be added as the validation progresses.

a. Import application server Certificate into F5

Import the certificates into F5 so the Server SSL profile can validate the application endpoint during SSL handshake (required due to self-signed/private CA cert).

- i. Retrieve the application server SSL certificate using **openssl** and copy the full certificate block (BEGIN to END).

Shell

```
openssl s_client -connect pvcbase-master.cldrsetup.local:8443 < /dev/null | sed -ne '/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p'
```

“Pvcbase-master.cldrsetup.local” here is the hostname of the application server e.g knox, NiFi.

Copy the full block from **BEGIN CERTIFICATE** to **END CERTIFICATE**.

ii. Import Certificate into F5

1. Open a web browser and go to the BIG-IP Configuration utility, using the public IP address on the management network, for example: <https://<external-ip-address>>.
2. On the Main tab, click **System → Certificate Management → Traffic Certificate Management → SSL Certificate List → Import**

Import Type: Certificate

Certificate Name: e.g., **knox_server_cert**

Paste certificate text

3. Click **Import**
4. Post import, you can view the certificate in the list.

Hostname: i-0ed1e144a2fc859e8.us-east-2.compute.internal | Date: Oct 17, 2025 | User: admin
IP Address: 10.0.0.200 | Time: 5:07 AM (PDT) | Role: Administrator

ONLINE (ACTIVE)
Standalone

Main | Help | About

System » Certificate Management : Traffic Certificate Management : SSL Certificate List » **knox_certificate**

Statistics | iApps | DNS | Local Traffic | Acceleration | Device Management | Shared Objects | Security | Network

System

- Configuration
- File Management
- Certificate Management
- Disk Management
- Software Management
- License
- Resource Provisioning
- Platform
- High Availability
- Archives
- Services
- Preferences

General Properties

Name	knox_certificate
Partition / Path	Common
Certificate Subject(s)	cdpar-725-sdx-01.cldr.internal

Certificate Properties

Public Key Type	RSA
Public Key Size	3072 bits
Expires	May 22 2026 16:59:59 PDT
Version	1
Serial Number	a5:40:ab:85:f6:13:da:b3
Fingerprint	SHA256/A4:4A:3F:AB:00:7F:68:92:D7:78:A8:E4:7D:F3:82:A7:54:06:22:86:96:68:EB:32:32:5A:30:39:B8:18:B6:2B
Subject	Common Name: cdpar-725-sdx-01.cldr.internal Organization: Division: Locality: State Or Province: CA Country: US
Issuer	Common Name: SCM Local CA on cdpar-725-cmhost-01.cldr.internal Organizational Unit: Division: Locality: State Or Province: CA Country: US
Email	
Subject Alternative Name	DNS:cdpar-725-sdx-01.cldr.internal

Monitoring Properties

Monitoring Type	☒ OSCP
Issuer Certificate	None
OCSP	None

Hostname: i-0ed1e144a2fc859e8.us-east-2.compute.internal | Date: Oct 17, 2025 | User: admin
IP Address: 10.0.0.200 | Time: 5:07 AM (PDT) | Role: Administrator

ONLINE (ACTIVE)
Standalone

Main | Help | About

System » Certificate Management : Traffic Certificate Management : SSL Certificate List » **knox_certificate**

Statistics | iApps | DNS | Local Traffic | Acceleration | Device Management | Shared Objects | Security | Network

Key Properties

Name	knox_certificate
Partition / Path	Common
Key Type	No key

5. Similarly import NiFi server certificates into the F5.

b. Create Server SSL Profile

Create a Server SSL profile to validate the backend certificate.

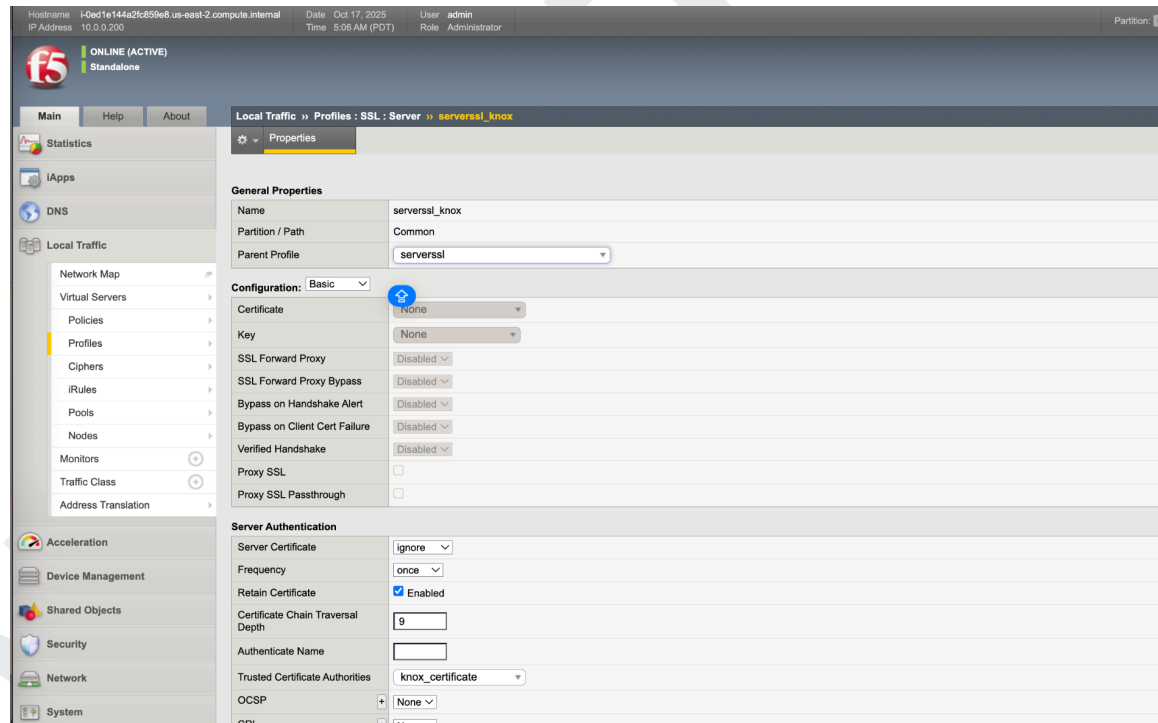
- i. On the Main tab, click **Local Traffic** → **Profiles** → **SSL** → **Server** → **Create**

1. Name: **serverssl_knox**

2. Under Server Authentication

Update **Trusted Certificate Authorities**: **knox_server_cert**

- ii. Click **Finished**



Similarly create a NiFi server SSL profile.

c. Create iRule

Create an iRule to rewrite the Host header and/or URL path as required by Knox, NiFi rerouting;

i. On the main tab, Click **Local Traffic** → **iRules** → **iRule List** → **Create**

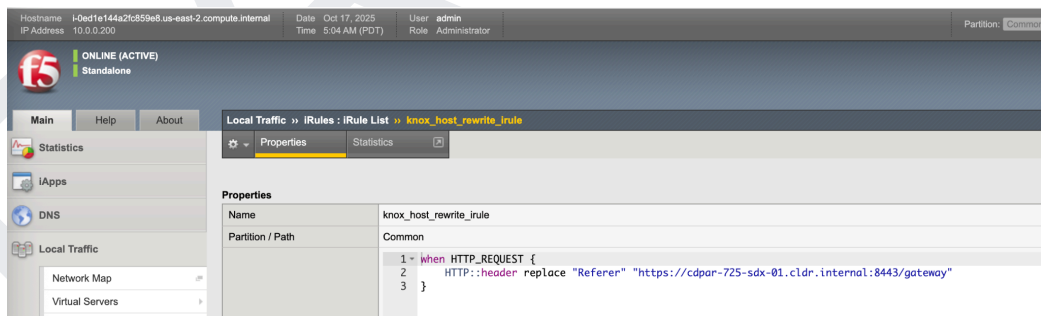
1. Knox

- **Name:** knox_host_rewrite_irule
- Definition:

None

```
when HTTP_REQUEST {  
    HTTP::header replace "Referer"  
    "https://cdpar-725-sdx-01.cldr.internal:8443/gateway"  
}
```

Where “<https://cdpar-725-sdx-01.cldr.internal:8443/gateway>” is the UI access url for knox.



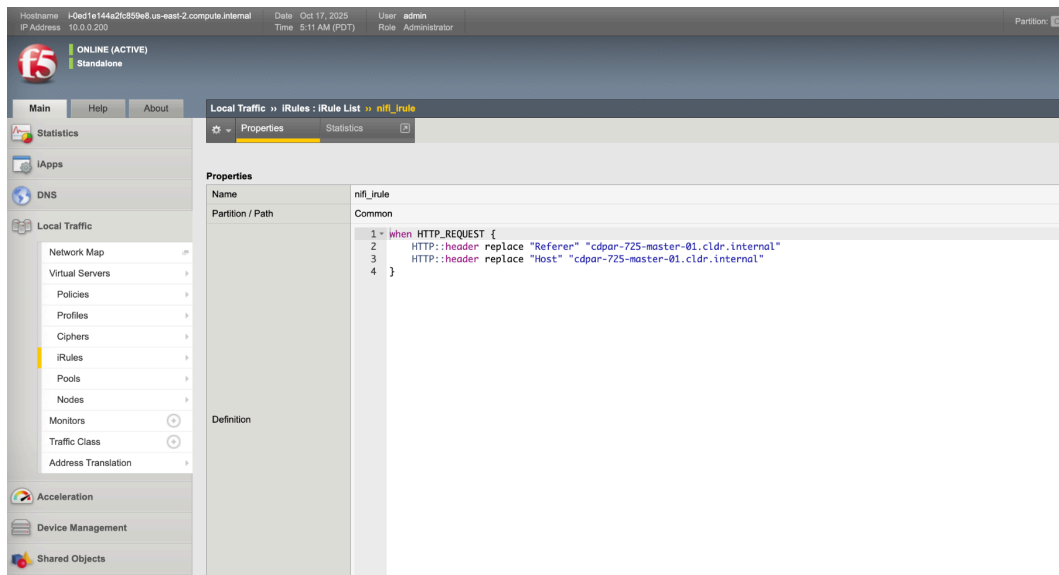
2. NiFi

- **Name:** knox_host_rewrite_irule
- Definition:

None

```
when HTTP_REQUEST {  
    HTTP::header replace "Referer" "cdpar-725-master-01.cldr.internal"  
    HTTP::header replace "Host" "cdpar-725-master-01.cldr.internal"  
}
```

Where “cdpar-725-master-01.cldr.internal” is the server hostname for NiFi.



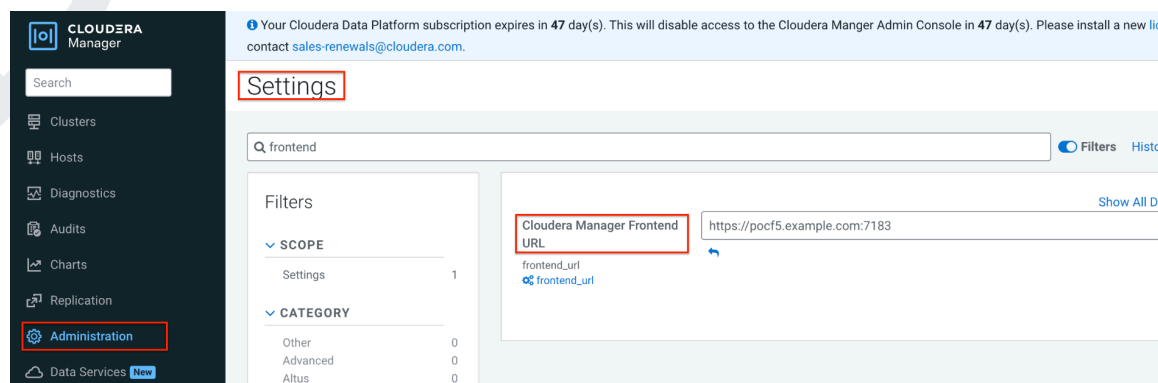
d. Cloudera Manager Access – Reverse Proxy / Load Balancer in Front

When Cloudera Manager is installed **without** a proxy and an F5 VIP is introduced later, CM must be explicitly configured to recognize that it is now accessed through a reverse proxy.

In such deployments, CM UI must use the **F5 external URL** instead of the internal Cloudera hostname.

Required Configuration Changes in Cloudera Manager:

- i. Set the Fronted URL of Cloudera Manager
Goto CM UI > Administration > Settings > Search for “frontend” > update the **frontend_url** field
Value: <https://pocf5.example.com:7183>



- ii. Make CM Trust Proxy Headers (Critical)

1. Edit the file: `/etc/default/cloudera-scm-server`
2. Locate the existing line containing
 `"CMF_JAVA_OPTS"`
3. Add this below it:

Shell

```
export CMF_JAVA_OPTS="${CMF_JAVA_OPTS}  
-Dcom.cloudera.cmf.web.forwarded.headers.enabled=true"
```

```
#  
# Java Options.  
#  
# Default value sets Java maximum heap size to 8GB  
#  
export CMF_JAVA_OPTS="-Xmx8G -XX:+HeapDumpOnOutOfMemoryError -XX:HeapDumpPath=/tmp -Dcom.sun.management.jmxremote.ssl.enabled.protocols=TLSv1.2 -Dorg.apache.avro.specific.use_custom_coders=true"  
export CMF_JAVA_OPTS="${CMF_JAVA_OPTS} -Dcom.cloudera.cmf.web.forwarded.headers.enabled=true"
```

iii. Restart Cloudera Manager

This is required because the change is a JVM startup flag.

systemctl restart cloudera-scm-server

e. Knox Access – Reverse Proxy / Load Balancer in Front

When **Apache Knox is accessed through an upstream reverse proxy or load balancer (for example, F5 BIG-IP)**, Knox must be explicitly configured to recognize the external endpoint.

If this is not done, users may encounter **Knox Access Exception** errors due to incorrect redirect URLs or host validation failures.

In such deployments, Knox must use the **F5 external URL** instead of the internal Cloudera hostname.

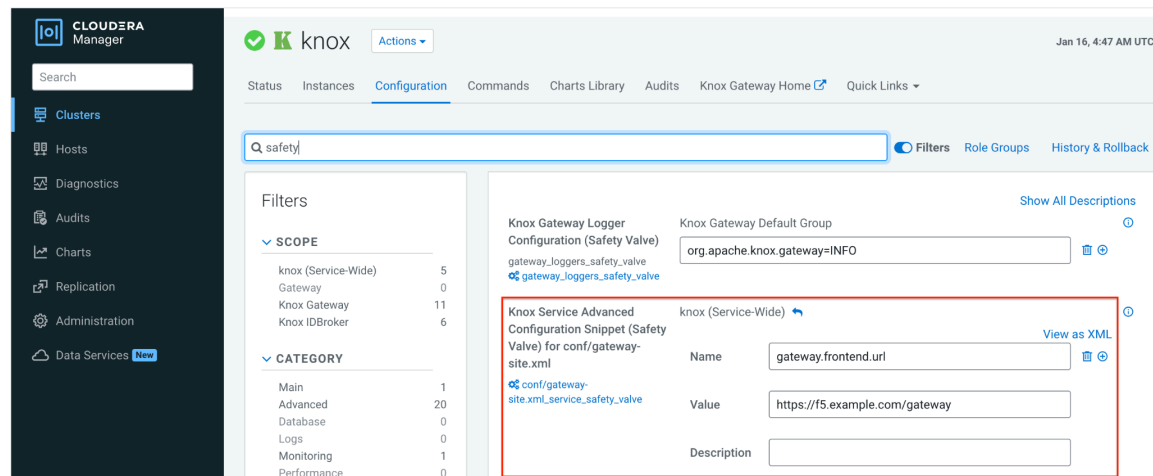
Required Configuration Changes in Cloudera Manager:

i. **Update Knox Safety Valve for Topology / Gateway**

Goto CM UI > Knox > Configuration > Search for “Safety” > update the **conf/gateway-site.xml_service_safety_valve** field

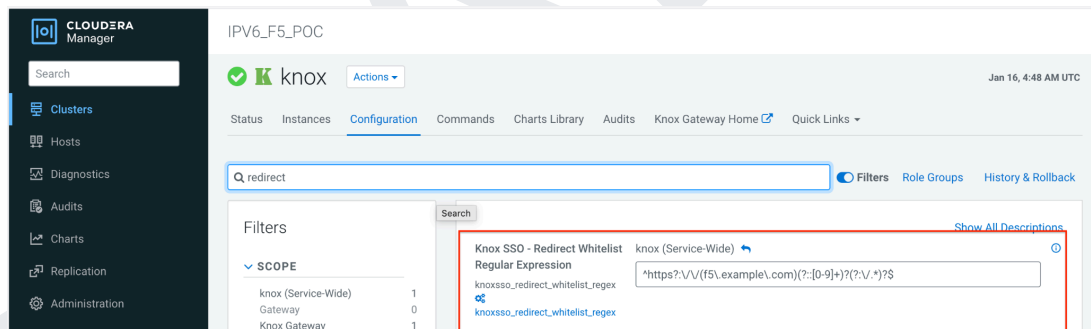
Name: gateway.frontend.url

Value: `https://f5.example.com/gateway`



ii. Update Knox Redirect URL

Goto CM UI > Knox > Configuration > Search for “redirect” > update the **knoxsso_redirect_whitelist_regex** field
Value: `^https?:VV(f5\.example\.com)(?:[0-9]+)?(?:V.*)?$`



8. Create a virtual server to access different applications

a. Cloudera Manager

- In the BIG-IP Configuration utility, on the Main tab, click **Local Traffic** -> **Virtual Servers**.
- Click **Create** and populate the following fields.

Field	Value
Name	A unique name
Source Address	:::/0 for ipv6

Destination Address/Mask	10.0.1.202 – for ipv4 routing 2600:1f16:123:601:0:0:0:202 – for ipv6 routing
Service Port	A port number or a service name from the Service Port list. e.g 7183 (default for Cloudera Manager)
HTTP Profile	None
Source Address Translation	Auto Map
Default Pool	cm_pool (pool name created in previous step)

Hostname: i-0ed1e144a2fc859e8.us-east-2.compute.internal Date: Oct 17, 2025 User: admin
IP Address: 10.0.0.200 Time: 5:08 AM (PDT) Role: Administrator

ONLINE (ACTIVE)
Standalone

Main Help About

Local Traffic » Virtual Servers : Virtual Server List » **cdp_pool**

Properties Resources Security Statistics

General Properties

Name	cdp_pool
Partition / Path	Common
Description	
Type	Standard
Source Address	☒ Host ☐ Address List ::/0
Destination Address/Mask	☒ Host ☐ Address List 2600:1f16:123:601:0:0:0:202
Service Port	☒ Port ☐ Port List 7183 Other:
Notify Status to Virtual Address	☒
Availability	☒ Unknown (Enabled) - The children pool member(s) either don't have service checking enabled, or service check results are not available
Synccookie Status	Inactive
State	Enabled

Configuration: Basic

Protocol	TCP
Protocol Profile (Client)	tcp
Protocol Profile (Server)	(Use Client Profile)
HTTP Profile (Client)	None
HTTP Profile (Server)	(Use Client Profile)
HTTP Proxy Connect Profile	None
FTP Profile	None
RTSP Profile	None
PPTP Profile	None

b. Hue

- i. Similarly for Hue, click **Local Traffic** –> **Virtual Servers**.
- ii. Click **Create** and populate the following fields.

Field	Value
Name	A unique name
Source Address	::/0 for ipv6
Destination Address/Mask	10.0.1.202 - for ipv4 routing 2600:1f16:123:601:0:0:0:202 - for ipv6 routing
Service Port	A port number or a service name from the Service Port list. e.g 8888 (default for Hue)
HTTP Profile	None
Source Address Translation	Auto Map
Default Pool	hue_pool (pool name created in previous step)

c. Knox

- i. For Knox, click **Local Traffic -> Virtual Servers**.
- ii. Click **Create** and populate the following fields.

Field	Value
Name	A unique name
Source Address	::/0 for ipv6
Destination Address/Mask	10.0.1.202 - for ipv4 routing 2600:1f16:123:601:0:0:0:202 - for ipv6 routing
Service Port	A port number or a service name from the Service Port list. e.g 8444
HTTP Profile	http
Source Address Translation	Auto Map
SSL Profile (Client)	clientssl
SSL Profile (Server)	serverssl_knox
Default Pool	knox_pool (pool name created in previous step)
iRules	knox_host_rewrite_irule

Configuration: Basic

Protocol	TCP
Protocol Profile (Client)	tcp
Protocol Profile (Server)	(Use Client Profile)
HTTP Profile (Client)	http
HTTP Profile (Server)	(Use Client Profile)
HTTP Proxy Connect Profile	None
FTP Profile	None
RTSP Profile	None
PPTP Profile	None

SSL Profile (Client)	Selected /Common clientssl	Available /Common clientssl-insecure-compatible clientssl-quic clientssl-secure cm_certificate_clientssl cm_clientssl crypto-server-default-clientssl
SSL Profile (Server)	Selected /Common serverssl_knox	Available /Common apm-default-serverssl cloud-service-default-ssl cm_serverssl crypto-client-default-serverssl do-not-remove-without-replacement f5aas-default-ssl

SMTPS Profile	None
POP3 Profile	None
Client LDAP Profile	None
Server LDAP Profile	None
Service Profile	None
SMTP Profile	None
VLAN and Tunnel Traffic	All VLANs and Tunnels
Source Address Translation	Auto Map

Resources

iRules	Enabled /Common knox_host_rewrite_irule Up Down	Available _sys_auth_tacacs _sys_https_redirect cdp_irule cdp_snirule nifi_irule
Policies	Enabled	Available
Default Pool	+ knox_pool	
Default Persistence Profile	None	
Fallback Persistence Profile	None	

Cancel
Repeat
Finished

d. Nifi

- Similar to **knox**, In the BIG-IP Configuration utility, on the Main tab, click **Local Traffic -> Virtual Servers**.
- Click **Create** and populate the following fields.

Field	Value
Name	A unique name
Source Address	::/0 for ipv6
Destination Address/Mask	10.0.1.202 - for lpv4 routing 2600:1f16:123:601:0:0:0:202 - for ipv6 routing
Service Port	A port number or a service name from the Service Port list. e.g 8443 (default for Nifi)

HTTP Profile	http
Source Address Translation	Auto Map
SSL Profile (Client)	clientssl
SSL Profile (Server)	serverssl_nifi
Default Pool	nifi_pool (pool name created in previous step)
iRules	nifi_irule

e. ECS Control Plane

- i. Similar to **CM & Hue**, In the BIG-IP Configuration utility, on the Main tab, click **Local Traffic -> Virtual Servers**.
- ii. Click **Create** and populate the following fields.

Field	Value
Name	A unique name
Source Address	::/0 for ipv6
Destination Address/Mask	10.0.1.202 - for ipv4 routing 2600:1f16:123:601:0:0:0:202 - for ipv6 routing
Service Port	443
HTTP Profile	None
Source Address Translation	Auto Map
Default Pool	ecs_pool (pool name created in previous step)

Phase 5: Accessing different applications via F5

Applications published through F5 BIG-IP are accessed using the DNS name mapped to the F5 **external virtual server IP (IPv4/IPv6)**, along with the **port and application-specific path** configured on the virtual server..

There are two supported access methods depending on the environment.

1. Using DNS (Recommended for Production)

Specific DNS configuration is required.

2. Using Local Host Mapping (Testing / POC Only)

a. Edit the local **hosts** file:

- i. Linux/macOS: **/etc/hosts**
- ii. Windows: **C:\Windows\System32\drivers\etc\hosts**

b. Add entries for both IPv4 and/or IPv6 as required. You can add the ECS Control plane fqdn or any other application running on top of ECS. Also add a new fqdn for f5 external ip to access base applications.

Shell

```
<IPv4-address> f5.example.com [<prefix>.<clusterdomain>]  
<IPv6-address> f5.example.com [<prefix>.<clusterdomain>]
```

```
##f5POC  
  
IPv4  
100.52.101.169 service.cde-l55pkxmz.apps.ecs.34.228.204.62.pvc.cloudera-labs.com hlzs5lx2.cde-l55pkxmz.apps.ecs.34.228.204.62.pvc.cloudera-labs.com pocf5.example.com hue-pocf5-impala.apps.ecs.34.228.204.62.pvc.cloudera-labs.com console-cdp.apps.ecs.34.228.204.62.pvc.cloudera-labs.com viz-pocf5-dataviz.apps.ecs.34.228.204.62.pvc.cloudera-labs.com  
  
IPv6  
2600:1f18:307d:8201::200 hue-pocf5-impala.apps.ecs.34.228.204.62.pvc.cloudera-labs.com viz-pocf5-dataviz.apps.ecs.34.228.204.62.pvc.cloudera-labs.com console-cdp.apps.ecs.34.228.204.62.pvc.cloudera-labs.com pocf5.example.com
```

The second method has been used for this POC. And the applications can be accessed with below URIs.

Shell

```
Cloudera Manager: https://f5.example.com:7183/cm/   
Knox: https://f5.example.com:443/gateway/homepage/home   
NiFi: https://f5.example.com:8443/nifi/login   
Hue: https://f5.example.com:8888/hue
```

Note: Make sure to whitelist your ip against **VIP** Ports on **VirtualServer** security group before accessing the applications via f5.

POC Environment Details

A total of 18 nodes were deployed for this POC, consisting of one BIG-IP instance and 17 Cloudera nodes.

While the Cloudera environment was provisioned via Goes automation—which includes a reverse proxy on the gateway node for backend access—this proxy was disabled and remained unused throughout the POC.

1. Server Details:

Name	Instance type	Hostname	Groups
pocf5-base-master-01	t3a.xlarge	pocf5-base-master-01.cldr.internal	base, base_masters, cluster,
pocf5-base-master-02	t3a.xlarge	pocf5-base-master-02.cldr.internal	base, base_masters, cluster
pocf5-base-master-03	t3a.xlarge	pocf5-base-master-03.cldr.internal	base, base_masters, cluster
pocf5-base-worker-01	t3a.xlarge	pocf5-base-worker-01.cldr.internal	base, base_workers, cluster, deployment, proxied_servers
pocf5-base-worker-02	t3a.xlarge	pocf5-base-worker-02.cldr.internal	base, base_workers, cluster, deployment, proxied_servers
pocf5-base-worker-03	t3a.xlarge	pocf5-base-worker-03.cldr.internal	base, base_workers, cluster, deployment, proxied_servers
pocf5-base-worker-04	t3a.xlarge	pocf5-base-worker-	base,

		04.cldr.internal	base_workers, cluster, deployment, proxied_servers
pocf5-ecs-master-01	r5a.4xlarge	pocf5-ecs-master-01.cldr.internal	cluster, deployment, ecs, ecs_masters, proxied_servers
pocf5-ecs-worker-01	r5a.4xlarge	pocf5-ecs-worker-01.cldr.internal	cluster, deployment, ecs, ecs_workers, proxied_servers
pocf5-ecs-worker-02	r5a.4xlarge	pocf5-ecs-worker-02.cldr.internal	cluster, deployment, ecs, ecs_workers, proxied_servers
pocf5-ecs-worker-03	r5a.4xlarge	pocf5-ecs-worker-03.cldr.internal	cluster, deployment, ecs, ecs_workers, proxied_servers
pocf5-ecs-worker-04	r5a.4xlarge	pocf5-ecs-worker-04.cldr.internal	cluster, deployment, ecs, ecs_workers, proxied_servers
pocf5-ecs-worker-05	r5a.4xlarge	pocf5-ecs-worker-05.cldr.internal	cluster, deployment, ecs, ecs_workers, proxied_servers
pocf5-gateway-01	t3a.medium	pocf5-gateway-01.cldr.internal	reverse_proxy, ssh_proxy
pocf5-manager-01	r5a.xlarge	pocf5-manager-01.cldr.internal	cloudera_manager, deployment, proxied_servers
pocf5-sdx-01	t3a.xlarge	pocf5-sdx-01.cldr.i	base, cluster,

		nternal	deployment, knox_gateway, proxied_servers, sdx
pocf5-services-01	t3a.large	pocf5-services-01. cldr.internal	freeipa, pgadmin, postgresql, proxied_servers
BigIpVM	m5.xlarge		

2. Version Details:

Software Component	Version
Cloudera Manager (CM)	7.13.1
Base Version	7.3.1
Data Services (DS)	1.5.5
OS	Red Hat 9.4
Cloudera Data Engineering (CDE)	1.24.0-h1-b10
Cloudera Data Warehouse (CDW)	1.11.0-h1-b1
Cloudera Data Visualization	7.2.7-b48 (on CDW) / 8.0.3 (on Base)
HUE	2025.0.19.3-2 (on CDW) / cdh7.3.1.401 (on Base)
Cloudera Flow Management (CFM)	2.1.7.3000